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INFORMATICS PLATFORM

Sapio Hosting Options Guide

Sapio offers several data hosting options for our customers. We offer these hosting options to effectively transfer the responsibility of managing the software and hardware environments necessary to support the Sapio Informatics Platform from our customers' internal IT teams to Sapio.

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Sapio's Hosting Options Include

Hosting in Sapio can be broken into two overarching classes : Multi-Tenant SaaS or Dedicated Hosting.

Multi-Tenant SaaS Editions

Multi-Tenant SaaS Editions are called Sapio Bronze, Silver, and Gold. These editions offer different feature sets to address varying customer needs.

- Customers share the instance with other customers, but all data is delineated in separate schemas
- Sapio will upgrade all instances at its discretion, but with sufficient notice to client with dev environments being updated first

Dedicated Hosting

Dedicated Hosting instances are called Dedicated SaaS, On-Prem, or Platinum. These instances are used when some or all data must be separate from the Multi-tenant SaaS offering.

- The customer does not share the instance with any other customers
- Sapio maintains the instance for the customer, essentially acting as the IT department for our clients by doing the initial installation, active monitoring, and performing upgrades
- Upgrades are elective, but recommended

See the summary table below to determine which offering best suits your needs

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sapio sciences

Sapio Hosting Options Comparison

Key
Sapio Controlled
Customer Controlled

	Multi-Tenant SaaS	Dedicated SaaS	On-Prem
Application Server	Shared servers logically segregated for each customer Cloud based, hosted on AWS	Customer-specific server Cloud based, hosted on AWS	Locally hosted on customer hardware
Database Server	Sapio hosted and managed Data logically segregated on servers shared with other customers by region	Sapio hosted and managed Data physically segregated	Customer hosted and manged All data separate from other customers
Hosting Fee	Included in intial purchase	Separate Add-on	N/A
Hosting Management Performed by	Sapio	Sapio	Customer
Upgrade Schedule	Sapio defined (weekends and after hours)	Customer chooses when to upgrade	Customer chooses when to upgrade
Upgrade Cadence	Timed with Releases 1 week between deployment to Dev, Test/QA, and Production environments	Customer chooses when to upgrade	Customer chooses when to upgrade
Platform Version Support	Always current	2 years from Release	2 years from Release
Site-to-Site VPN to VPC	No	No	N/A

*Sapio Platinum (not listed above) is no longer a part of our product hosting offering. Clients with existing, active Platinum licenses continue to be supported.

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Sapio Amazon Web Service Hosting Details for SaaS Users (Multi-Tenant/Dedicated)

Sapio Sciences uses Amazon Web Services (AWS) as a Hosting Partner and leverages several elements of AWS as part of its standard deployment. AWS data is hosted in data centers worldwide with deployment and monitoring tools that allow new environments to be spun up rapidly. The main AWS elements that are part of standard deployment are listed below : Amazon S3 Storage, AWS Elastic Cloud Computing (EC2), Amazon Elastic Block Store (EBS), Amazon Relational Database Service (RDS).

Note : EC2/EBS, RDS and S3 are encrypted in-transit and at-rest

Amazon S3 Storage (S3)

Secure, scalable, and durable file storage infrastructure designed for mission-critical primary data storage. The Amazon S3 Service Level Agreement (SLA) provides 99.99% availability and 99.999999999% durability (average annual expected loss of 0.000000001% of objects).

Unstructured data (PDFs, Office documents, and Instrument files) stored in S3 where data is encrypted. There are also multiple availability zones with unlimited versioning enabled for data restoration.

AWS Elastic Cloud Computing (EC2)

EC2 provides on-demand, scalable computing capacity that leverages a highly available infrastructure with 99.9% availability while eliminating the need to invest in hardware. Amazon EC2 enables users to add virtual servers as needed, configure security or networking, and manage storage. EC2 also uses daily restore snapshots that are retained for 14 days.

Amazon Elastic Block Store (EBS)

EBS is an easy-to-use, high-performance block-storage service designed for use with EC2. The Amazon EBS SLA that provides 99.9% availability. Data is AES-256 encrypted.

Amazon Relational Database Service (RDS)

RDS provides cost-efficient and resizable capacity while automating time-consuming administration tasks such as hardware provisioning, database setup, patching and backups. Multi-availability zones are utilized to provide 99.95% availability with synchronous replication to other availability zones within the region.

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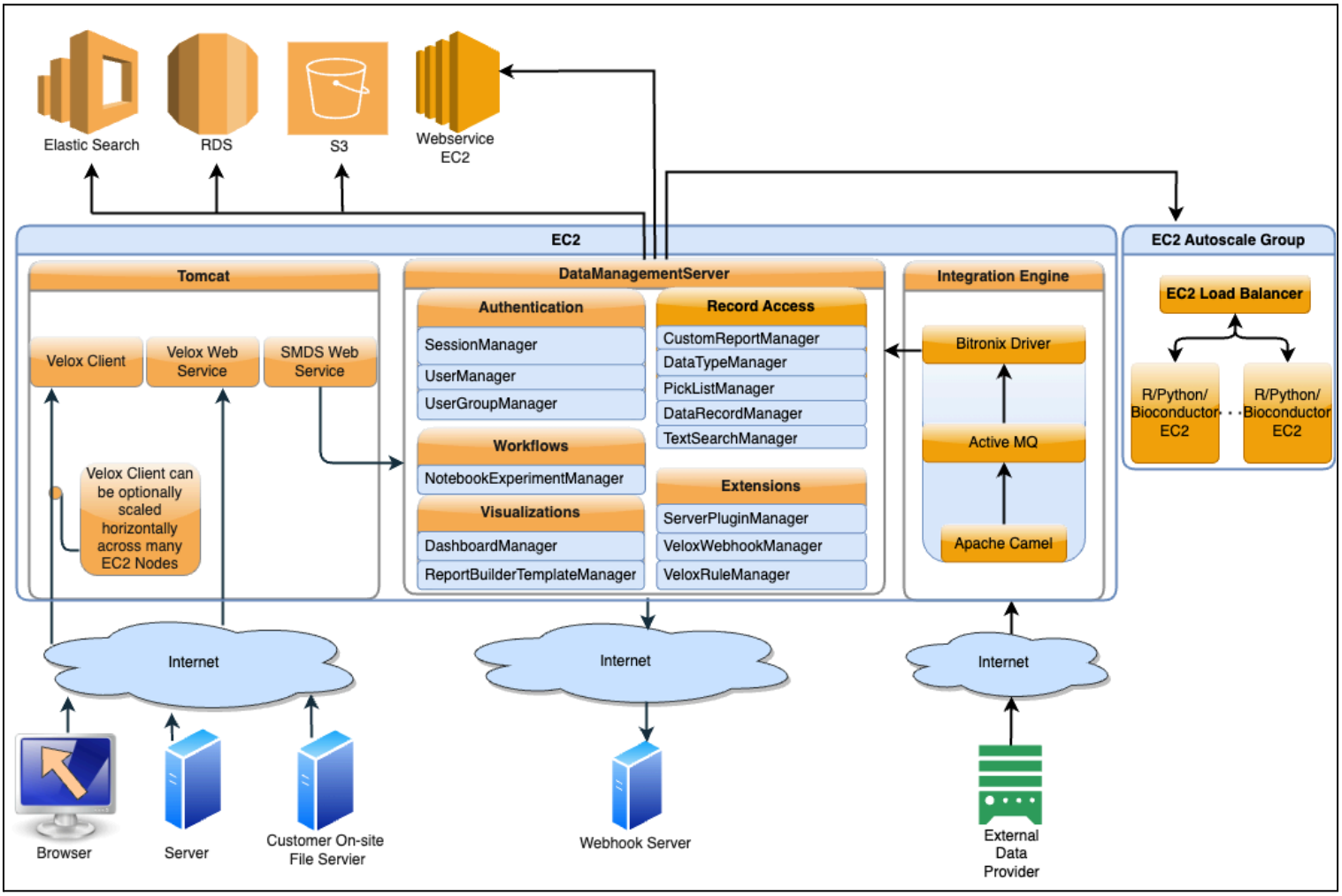
Structured data (tables of entities composed of fields) are stored in RDS where data is encrypted. There are also point-in-time restore snapshots that are retained for 14 days and full nightly backups retained for minimum 30 days.



Network Access Restrictions

Network access is restricted by IP address and SSH access limited to restricted personnel.

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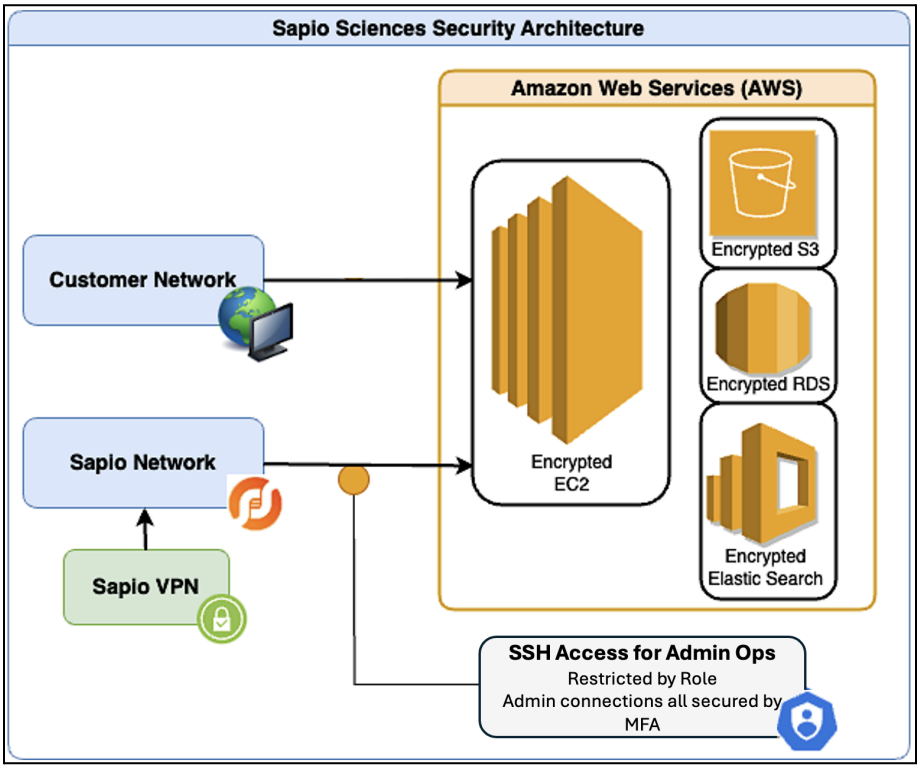
Security Architecture for SaaS Users (Multi-Tenant/Dedicated)

Sapio Contains the following security architecture for all environments, both Multi-Tenant and Dedicated...



...with the following features.

- Commitment to information security best practices demonstrated by adherence to:
  - SOC 2 Type 2
  - ISO 27001:2013
- Role-based access control
- Field and record-level access control
- Immutable audit log of data changes, system events, user access, and configuration changes
- Single-sign-on support

Next-gen Anti-malware software installed on all EC2 instances that includes 24x7 monitoring and remediation

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Sapio Hosting Details for On-Prem Users

In addition to providing and managing the setup of the AWS infrastructure, Sapio will host the software for the customer on provisioned items.

As part of the provisioned items, Sapio will provide an RDS instance with 2 vCPU and 4GB of RAM.

Note : Larger instances would require additional fees for 4 vCPU's and 16GB of RAM

A single tenant production and single tenant test/QA instance running the Sapio Informatics Server on an Amazon EC2 Server will also be provided.

Customer Responsibilities for On-Prem Instances

The customer of dedicated environments provides all administrative services of the AWS infrastructure including :

- Initial installation of the Sapio software
- Upgrading the Sapio software when new releases are available. The timing of this will be coordinated with the client, but the instances should never be running unsupported versions of Sapio software

Sapio Responsibilities for On-Prem Instances

Sapio will provide adequate lead time to customers with dedicated environments so that new releases can be tested before being applied to production. The only exception to updates are when the customer needs to patch a serious issue. Serious issues may require bypassing the normal test update protocol and patching production immediately with a limited notification.

Note : Product updates will attempt to be scheduled at the least disruptive times to customers and are expected to result in less than 1 hour of server down time

Sapio will coordinate with customers on the timing of deployment prior to updating all hosted environments. Additionally, Sapio will conduct daily backups to Amazon's S3 storage of the entire application definitions and data including the RDS database and stored items in S3 such as documents and images. The RDS database will also have a real time replicate with backups that are retained for 12 days max. Furthermore, both the EC2/EBS, RDS and S3 will be encrypted for data protection both in transit and while at rest.

Tip : Database and RDS snapshot backups are to another AWS region

Sapio also provides real time monitoring of the AWS components with alerts issued to Sapio personnel so that appropriate actions can be taken in a timely manner.

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